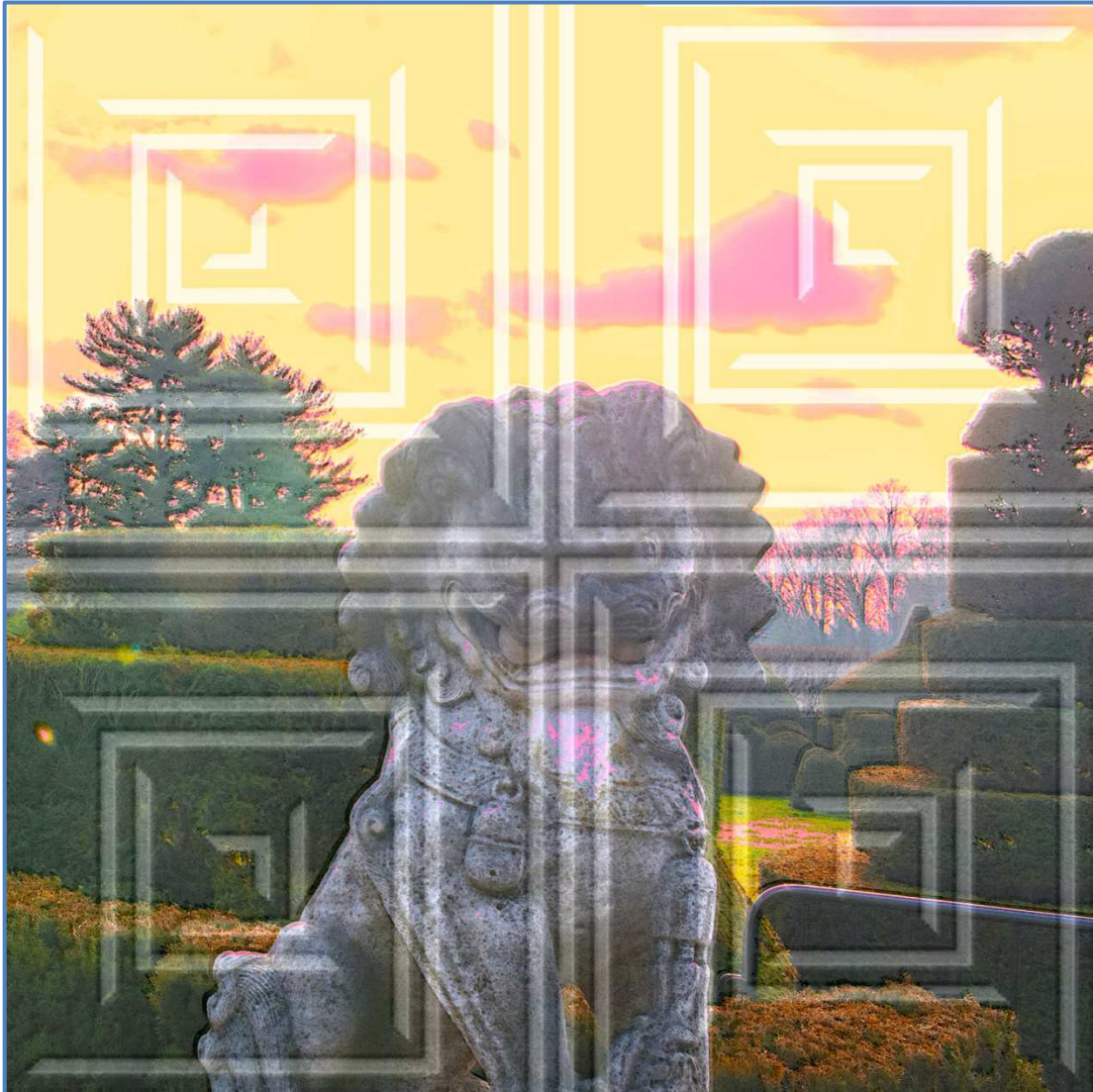


ChatGPT's Ontological Landscape



A Working Paper • November 1, 2023
William L. Benzon

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Abstract: Ontological concepts are the high-level category structures language. Physical objects can have certain attributes (form, texture, color, weight) and play certain roles with respect to verbs (they can be touched, hit, dropped, etc.). Living objects have those capacities, plus others as well. They can live, die, and grow. Animals have the capacities and affordances of plants, plus others than plants lack. And so it is with humans in respect to animals. This type of category structure is pervasive. Then we have the difference between concrete things, specifiable by how they strike the senses and abstract things, which are not (necessarily) perceptible. Consider the difference between “salt,” which is conceptually concrete and “NaCl,” which is a physical thing that is conceptually abstract. This kind of difference is pervasive as well. The first two sections of this paper explore ChatGPT on those matters. The third turns to the parlor game 20 questions as a way of exploring its ability to negotiate these conceptual structures. In games with six target concepts, each tried twice for a total of 12 games, ChatGPT performs better with abstract than with concrete targets.

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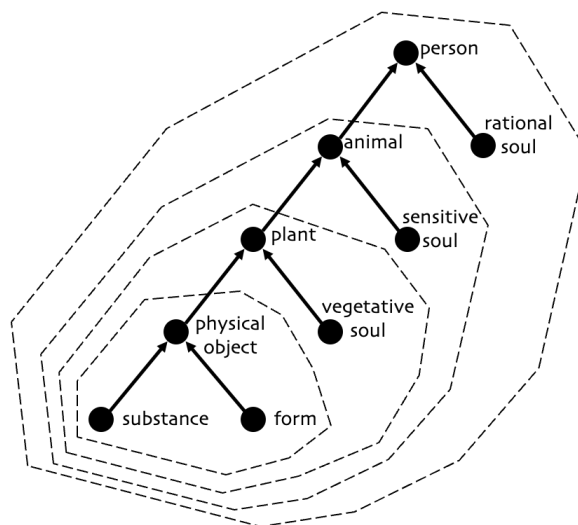
Introduction: Ontological structure in cognition

Perhaps the most famous sentence in twentieth century linguistics is one that Noam Chomsky made-up to argue that syntax is independent of semantics: “Colorless green ideas sleep furiously.” The syntax is fine, but the sentence is nonsense. Ideas are not the sorts of things that have colors or that can sleep. Those are what philosophers have come to call *category mistakes*.¹ Interestingly enough, the first time I asked ChatGPT to tell a story in which a colorless green idea was the protagonist, it refused:²

I'm sorry, but the concept of a "colorless green idea" is a nonsensical phrase that was invented as an example of a phrase that would be grammatically correct but semantically meaningless. It is not possible to create a story about a "colorless green idea" as it does not have any physical properties or characteristics that can be used in a story.

That’s one aspect of ontological cognition, and it is pervasive. Physical objects have form – cube, sphere, and so forth, but more complex forms are common – and are constituted of some substance or substances – stone, metal, wood, cloth, and so forth. They have textures (smooth, rough, slick, sticky, etc.) and colors (of which there are thousands). But they can’t shrink or grow. Plants can do those things, and reproduce as well. But they can’t move around of their own accord, nor do they have the capacity for sensation. Those capacities belong to animals, but animals cannot reason or speak. But humans can.

Those dependencies give rise to a conceptual structure that we can diagram like this:



¹ Magidor, Ofra, "Category Mistakes", *The Stanford Encyclopedia of Philosophy* (Fall 2022 Edition), Edward N. Zalta & Uri Nodelman (eds.), <https://plato.stanford.edu/archives/fall2022/entries/category-mistakes>.

² There’s truth, lies, and there’s ChatGPT [Realms of Being], New Savanna, Jan. 25, 2023, <https://new-savanna.blogspot.com/2023/01/theres-truth-lies-and-theres-chatgpt.html>.

I take the terminology of souls, vegetative, sensitive, and rational, from Aristotle, who laid it out in *De Anima*. Over time this structure became elaborated into a metaphysical trope called The Great Chain of Being.³ However, I'm not interested in the Chain as a philosophical idea, or an explicit conceptual trope for organizing the world. I'm interested in the perceptual and cognitive underpinnings of that structure; it is those that give rise to category mistakes and to ChatGPT's reluctance to tell a story with a colorless green idea as the protagonist.

This pervasive conceptual structure has another aspect that I like to illustrate with the difference between (ordinary table) *salt* and *NaCl*. Physically, they are pretty much the same thing, though salt has impurities that NaCl, by definition, does not. But conceptually they are quite different. Salt is defined by its color, the shape of its grains, and, above all, by its taste. One can sense the presence of salt by taste alone. This is a concept readily available to childrens and even infants, depending on just what you mean by "concept." NaCl is quite different. It exists in a conceptual matrix that didn't exist until the nineteenth century, one involving chemical elements, atoms, molecules, and bonds between atoms. The same difference exists between *water* and *H₂O* and *laughing gas* and *N₂O*, and so on for a whole variety of common substances. Nineteenth century chemistry imposes a conceptual structure on the world that is different from that of common-sense.

And this difference, between common-sense conceptualizations, and more sophisticated ones, science if you will, though not only science-proper, is pervasive. Biologists have Latinate names for creatures you and I see as rose bushes, pine trees, dogs, cats, and lobsters. So it goes with physicists, geologists, astronomers, even sociologists, economists and other social scientists, not to mention – *would you believe it?* – literary critics. Our conceptual universe is filled with specialized conceptual repertoires belonging to specialized disciplines.

What does ChatGPT know of any of this? Well, it likely knows all the terms, certainly many more terms across many more disciplines than any one human being knows. Whether or not it understands those terms deeply, or only superficially, those are other questions, well beyond the scope of this paper.

From Salt and NaCl to the Great Chain – I quiz ChatGPT on the basics, salt vs. NaCl, Morning Star and Evening Star, The Great Chain of Being, *De Anima*, Plato's parable of the charioteer, and the study of ontology.

Mapping ChatGPT's ontological landscape, gradients and choices [interpretability] – I start by exploring the difference between concrete and abstract questions, move on to the Great Chain, and conclude by asking it how 20 questions exploits the semantic structure of language. Along the way I introduce Waddington's epigenetic landscape as a metaphor for ChatGPT's operation.

³ Wikipedia, Great Chain of Being, https://en.wikipedia.org/wiki/Great_chain_of_being.
Nee, S., The great chain of being, *Nature* **435**, 429 (2005), <https://doi.org/10.1038/435429a>.

ChatGPT Plays 20 Questions [& sometimes needs help] – I explore ChatGPT’s facility with categories by playing 20 questions with it, two rounds each on: bicycle, squid, justice, apple, evolution, and truth. On the whole it seemed to do better with abstract concepts than with concrete.

Appendix: Some papers on cognitive ontology – Four papers, including abstracts. Three are specifically about conceptual ontology. The fourth is about cultural evolution and discusses the elaboration of conceptual ontology over the long course of cultural evolution, *salt* vs. *NaCl* writ large.

From Salt and NaCl to the Great Chain

This is all basic stuff. ChatGPT knows that salt and NaCl are the same, but it can also be prompted into explaining that table salt isn't pure NaCl. I then ran it through the Morning Star and the Evening Star on the way to the Great Chain, which it explicated reasonably well. I ended on the topic of ontology itself.

Salt & NaCl

As I've indicated in the introduction, my point is that "salt" and "NaCl" are embedded in different conceptual ontologies. "Salt" is a common-sense term intelligible to young children while "NaCl" is from a scientific ontology that didn't emerge until the nineteenth-century. While ChatGPT does recognize a difference in meaning between the two terms, I doubt that that recognition is very deep. After all, it has no access to the physical world, it can't see or touch salt, much less taste it. As for NaCl, that conceptualization is constructed over laboratory observations and measurements, and mathematical calculations and concepts. With no access to the physical world, ChatGPT cannot make laboratory observations, though no doubt it could discourse on them if asked to do so, which I did not.

It did not, however, have any difficulty explaining that table salt and NaCl are not quite the same when I put the question to it: "Is table salt pure NaCl?" I firmed up its sense of the distinction with a follow-up question: "So table salt and NaCl aren't precisely the same thing?" *No, they are not*, it said in so many words. But that's a long way from understanding that *salt* and *NaCl* are embedded in different conceptual systems.

Here's the dialog:

Tell me what salt is?

Salt is a mineral substance composed primarily of sodium chloride (NaCl), a chemical compound belonging to the larger class of ionic salts. It is essential for the proper functioning of the human body, and is used in many different ways in cooking and food preparation.

Salt occurs naturally in many parts of the world, and is produced by the evaporation of seawater or by mining rock salt deposits. It is used to flavor and preserve food, and is also added to water in order to regulate the amount of chlorine and other chemicals that are used to treat it.

In addition to its use in food and water treatment, salt has a number of other uses. It is used in the production of various chemicals and in the manufacture of soap, glass, and textiles. It is also used in the de-icing of roads and the production of detergents and other cleaning agents.

What is NaCl?

Sodium chloride (NaCl) is a chemical compound that is made up of sodium and chlorine atoms. It is commonly known as table salt or just salt, and is used in many different ways in cooking and food preparation.

Sodium chloride is a white, crystalline solid that is highly soluble in water. It is an essential nutrient for the human body, and is used in the regulation of body fluids, nerve and muscle function, and blood pressure.

In addition to its use as a food seasoning and preservative, sodium chloride has a number of other uses. It is used in the production of various chemicals, such as sodium hydroxide and hydrochloric acid, and in the manufacture of soap, glass, and textiles. It is also used in the de-icing of roads and the production of detergents and other cleaning agents.

Are table salt and NaCl the same?

Yes, table salt and sodium chloride (NaCl) are the same thing. Table salt is simply a common name for sodium chloride, which is a chemical compound made up of sodium and chlorine atoms. It is a white, crystalline solid that is highly soluble in water and is used in many different ways in cooking and food preparation.

Table salt is an essential nutrient for the human body, and is used in the regulation of body fluids, nerve and muscle function, and blood pressure. It is also used as a food seasoning and preservative, and has a number of other uses, such as in the production of chemicals, the manufacture of soap, glass, and textiles, and in the de-icing of roads and the production of detergents and other cleaning agents.

Is table salt pure NaCl?

Table salt is mostly composed of sodium chloride (NaCl), but it may also contain other ingredients. Most table salt sold in the United States is iodized, which means that it has been fortified with iodine, an essential nutrient that helps to prevent thyroid problems. In addition to iodine, table salt may also contain other ingredients, such as anticaking agents, which help to prevent the salt from clumping together.

The purity of table salt can vary depending on the source and the processing method used. Some table salt is mined from underground deposits, while others are produced by the evaporation of seawater. Table salt that is mined from underground deposits may be more pure, while table salt produced by the evaporation of seawater may contain trace amounts of other minerals.

In general, however, table salt is considered to be relatively pure, with more than 97% of the weight of the salt being composed of sodium chloride.

So table salt and NaCl aren't precisely the same thing?

Table salt and sodium chloride (NaCl) are essentially the same thing, but table salt may contain other ingredients in addition to sodium chloride. Most table salt sold in the United States is iodized, which means that it has been fortified with iodine, an essential nutrient that helps to prevent thyroid problems. In addition to iodine, table salt may also contain other ingredients, such as anticaking agents, which help to prevent the salt from clumping together.

The purity of table salt can vary depending on the source and the processing method used. Some table salt is mined from underground deposits, while others are produced by the evaporation of seawater. Table salt that is mined from underground deposits may be more pure, while table salt produced by the evaporation of seawater may contain trace amounts of other minerals.

Overall, however, table salt is mostly composed of sodium chloride, with more than 97% of the weight of the salt being made up of this chemical compound.

Morning Star, Evening Star

This is a standard example that logicians use to illustrate the difference between intension and extension. The two terms have different intensions but the same extension.

Tell me about the Morning Star and the Evening Star.

The Morning Star and the Evening Star are two names that have been used to refer to the planet Venus when it appears in the sky. Venus is the brightest planet in the sky, and it is visible in the early morning hours before sunrise (the

Morning Star) and in the evening hours after sunset (the Evening Star).

Venus is often referred to as the Morning Star when it appears in the eastern sky just before sunrise, and as the Evening Star when it appears in the western sky just after sunset. These appearances are caused by Venus's orbit around the sun, which is closer to the sun than Earth's orbit. As a result, Venus appears to be much brighter than other planets and is often visible in the sky even when the sun is up.

Venus is sometimes also referred to as the "Earth's Sister Planet" because it is similar in size, mass, and composition to Earth. It is also the hottest planet in the solar system due to its thick atmosphere, which traps heat and causes a greenhouse effect.

Great Chain

Here I am asking it a set to questions related to the concept of the Great Chain of Being, starting with the Great Chain itself. Aristotle's *De Anima* provides the conceptual core. Plato's parable of the charioteer is related to the idea that the human soul is not unitary.

What is the Great Chain of Being?

The Great Chain of Being is a concept that was developed in ancient and medieval philosophy and was used to describe the natural order of the universe. It was believed that the universe was hierarchical, with each level of the hierarchy being made up of beings or substances that were ranked according to their complexity, perfection, and closeness to God.

According to the Great Chain of Being, the universe was divided into a number of levels, with the most simple and imperfect substances at the bottom and the most complex and perfect substances at the top. At the very top of the hierarchy was God, followed by angels and other celestial beings. The next level down consisted of humans, and then animals, plants, minerals, and finally inanimate objects.

The Great Chain of Being was used to explain the natural order of the universe and to justify the social and political hierarchy of human society. It was also used to explain why certain events occurred and why some people were more successful or fortunate than others. Despite its widespread influence, the concept of the Great Chain of Being has largely been abandoned by modern science and philosophy.

De Anima

According to Aristotle's De Anima, how many souls do people have?

According to Aristotle's De Anima (On the Soul), humans have a single soul that is divided into three parts: the rational part, the spirited part, and the appetitive part.

The rational part of the soul is responsible for reasoning and thought, and is what makes humans unique among all other living beings. The spirited part of the soul is responsible for emotions and desires, and is what separates humans from plants. The appetitive part of the soul is responsible for the desire for food, drink, and other physical pleasures.

Aristotle believed that the soul was the source of life and the driving force behind all living things. He believed that the soul was immortal and that it was the source of all thought, action, and sensation.

While Aristotle's ideas about the soul have had a lasting influence on Western thought and philosophy, they are not universally accepted and have been the subject of much debate and criticism.

What about animals?

According to Aristotle's De Anima (On the Soul), animals also have a soul, but their soul is not as complex as the human soul. Aristotle believed that the soul of an animal was made up of two parts: the vegetative part, which is responsible for the basic functions of life, such as nourishment, growth, and reproduction; and the sensitive part, which is responsible for perception and movement.

Aristotle believed that the soul of an animal was not immortal and that it did not have the capacity for reason or thought. He believed that animals were driven by their instincts and that they were incapable of rational thought or decision-making.

While Aristotle's ideas about the soul have had a lasting influence on Western thought and philosophy, they are not universally accepted and have been the subject of much debate and criticism. Modern scientific understanding of the nature of the mind and consciousness differs significantly from Aristotle's ideas, and many scientists and philosophers

reject the idea of a soul or an immaterial aspect of the mind.

Very good.

Thank you! I'm glad that I could help. If you have any other questions, feel free to ask. I'll do my best to help.

Tell me Plato's parable of the charioteer.

The parable of the charioteer is a metaphor used by Plato in his work the Phaedrus to describe the nature of the human soul and its relationship to the body.

In the parable, Plato compares the human soul to a chariot being pulled by two horses. The two horses represent the two parts of the soul: one is noble and good, and the other is base and violent. The charioteer represents reason, and it is his job to control the two horses and steer them in the right direction.

According to Plato, the noble horse represents the rational part of the soul, which is responsible for reason and contemplation. The base horse represents the passionate and appetitive part of the soul, which is driven by desire and pleasure. The charioteer must use reason to control the horses and keep them in balance, so that the soul can achieve its highest potential.

The parable of the charioteer is meant to illustrate the importance of reason in controlling the passions and desires of the soul and guiding it towards goodness and virtue. It is also meant to illustrate the tension between the rational and irrational aspects of the human soul and the need to find a balance between them.

Ontology

Now things get interesting. I asked it what ontology was and it gave me a standard, and correct, answer. Then I asked it: "Have we been discussing ontology? Explain." It said that, *no, we are not*. And that is true. I may have been asking it these questions with conceptual ontology in mind, but that's quite different from discussing the subject of conceptual ontology. That's a much heavier lift. I made a move in that direction by asking it about the the idea of ontology used in computer science and AI, to which it responded properly. But how do you get from there to the idea that our whole conceptual system is pervaded with ontological concepts, and thus that *salt* and *NaCl* exist in different conceptual ontologies? I've got to pause and take a breath to get my hands around that one myself, and I've been thinking about these issues for years.

Here's our discourse:

What is ontology?

Ontology is a branch of philosophy that deals with the nature of being or existence. It is concerned with the question of what exists and what it means for something to exist.

Ontology is closely related to metaphysics, which is the study of the nature of reality and the fundamental principles that govern the universe. While metaphysics is concerned with the nature of reality as a whole, ontology is more focused on the nature of specific types of entities and how they relate to one another.

Ontologists often ask questions such as: What is the nature of the universe? What are the fundamental categories of being? What are the essential properties of objects and entities? What is the relationship between mind and body? What is the nature of time and space?

Ontological inquiry has been a central concern of philosophers throughout history, and it has had a significant influence on many other fields, such as mathematics, logic, and computer science.

Have we been discussing ontology? Explain.

No, we have not been discussing ontology specifically. We have been discussing a variety of topics, including the nature of salt, the Great Chain of Being, and Plato's parable of the charioteer. While these topics may be related to ontological questions in some way, they do not directly address the nature of being or existence.

Ontology is a branch of philosophy that deals with the nature of being or existence and is concerned with questions such as: What is the nature of the universe? What are the fundamental categories of being? What are the essential properties of objects and entities? What is the relationship between mind and body? What is the nature of time and space?

We have not specifically addressed any of these ontological questions in our conversation. If you are interested in learning more about ontology or have specific questions about the nature of being or existence, I would be happy to try and help.

What does ontology mean in computer science and artificial intelligence?

In computer science and artificial intelligence, ontology refers to a systematic way of representing knowledge about a particular domain or subject area. An ontology is a formal representation of the concepts and relationships within a specific domain, and it is used to define the terms and categories that are used to describe the objects and concepts within that domain.

In the field of artificial intelligence, ontologies are used to represent knowledge about a particular domain in a way that can be understood by a computer. This allows the computer to reason about and make inferences about the domain using logical rules. Ontologies are used in a variety of applications, including natural language processing, information retrieval, and knowledge management.

Ontologies are typically represented using a standardized vocabulary and a set of rules for organizing and connecting the concepts and relationships within the domain. They can be created manually or automatically using machine learning techniques.

Overall, the use of ontologies in computer science and artificial intelligence helps to structure and organize knowledge about a particular domain in a way that can be understood and used by computers.

Mapping ChatGPT's ontological landscape, gradients and choices [interpretability]

By ontological landscape I mean the repertoire of different things it knows about. The parlor game, Twenty questions⁴, was invented in the United States in the 19th century as a way of playing around with and on this landscape. One player thinks of some phenomenon and the other has to guess it by asking a series of no more than 20 questions that have a “yes” or “no” answer. One popular variant of the game begins with a simple question: “Animal, vegetable, or mineral?” Once the question has been answered the questioner must continue with binary questions.

I'll play 20 questions with ChatGPT in the next section. In this section I take a look at how ChatGPT understands the distinction between abstract and concrete concepts and the relationships between them. I start simply, by asking it for lists of concrete and abstract objects, and move on from there. One of the things I'm looking for is just which things it lists, what does it repeat from one list to another? Then I ask it to name some abstract concepts that can be understood in terms of concrete concepts. Then I ask it to tell a story about hope, then some metaphors. Then I look at mechanical things then another look at The Great Chain of Being. I conclude by asking it to tell me about the game, 20 questions, and explain how it depends on the semantic structure of language.

All of the interactions reported below took place on the morning of October 15 using the September 25 version of ChatGPT.

Physical things

I decided to start with the simplest prompt I could imagine, “thing.” That's the most general category we've got, no? Anything whatever is going to be some kind of thing. But how would ChatGPT respond? It responded as I thought it would, so I gave it a more explicit prompt, to which it responded.

Thing

It seems like you mentioned "thing," but you didn't provide a specific question or context. Could you please clarify or provide more information so I can assist you better?

Give me a list of 20 things, anything.

Certainly! Here's a list of 20 random things:

⁴ Wikipedia, Twenty Questions, https://en.wikipedia.org/wiki/Twenty_questions.

1. A beach ball
2. A rainbow
3. A bicycle
4. A bookshelf
5. A coffee mug
6. A mountain
7. A laptop computer
8. A painting
9. A pineapple
10. A pair of sunglasses
11. A telescope
12. A running shoe
13. A wristwatch
14. A musical instrument
15. A flower vase
16. A hammock
17. A traffic light
18. A hot air balloon
19. A chef's knife
20. A Rubik's Cube

These are just a few examples of things, and the list could go on endlessly!

I had no particular expectations about how ChatGPT would respond. But I couldn't help but notice that each item in its response was some kind of physical thing. To be sure, rainbows (the second item in the list) are rather diaphanous, and you couldn't actually touch one (I should have asked ChatGPT about that; later...), but it is still a physical thing.

I didn't ask for physical things, but that's what ChatGPT gave me. That's worth noting.

Let's think a bit before continuing on. I wonder if that first list is "rigid" in the sense that ChatGPT would respond the same way given the same prompt. I suspect not, but it's easy enough to find out. I went back a bit later.

Give me a list of 20 things, anything.

Certainly! Here's a list of 20 random things:

1. Beach vacations
2. Chocolate chip cookies
3. Classic literature
4. Sunsets
5. Roller coasters
6. Coffee shops
7. Mountain hiking
8. Vintage cars
9. Starry nights

10. Watermelon
11. Jazz music
12. Sunflowers
13. Science fiction movies
14. Art galleries
15. Hot air balloons
16. Snow-capped mountains
17. Aquariums
18. Street food
19. Historical landmarks
20. Puzzle games

It's not the same list I got the first time, but almost all of them are physical things. Classic literature (#3) isn't really a physical thing, though books are. Mountain hiking (#7) is a physical activity, but whether or not you'd call it a physical thing is, well, an interesting issue. Jazz music (#7)? Science fiction movies (#13), puzzle games (#20)?

I decided to log out and then try again:

Give me a list of 20 things, anything.

Certainly! Here's a list of 20 random things:

1. Pencil
2. Bicycle
3. Sunflower
4. Smartphone
5. Coffee mug
6. Guitar
7. Rainbow
8. Lighthouse
9. Chocolate chip cookies
10. Telescope
11. Mountain
12. Soccer ball
13. Vintage typewriter
14. Starfish
15. Hot air balloon
16. Eiffel Tower
17. Polar bear
18. Sandcastle
19. Wristwatch
20. Lightning bolt

That's better. I notice we've got "rainbow" again (#7), and "Lightening bolt" (#20) is rather ephemeral as well, though if one were to hit you, the effect would be dramatic and quite physical. "Chocolate chip cookies" (#9) show up again, as does "Hot air balloon" and "Bicycle." These lists seem to be drawn from roughly the same conceptual territory. I

suppose that's not terribly surprising, but it certainly doesn't seem like ChatGPT is making random choices from among all available physical things. Why these things?

I decided to explicitly specify physical objects:

How about 20 physical objects.

Of course, here's a list of 20 physical objects:

1. Refrigerator
2. Sofa
3. Running shoes
4. Toaster
5. Sunglasses
6. Umbrella
7. Bookshelf
8. Camera
9. Toothbrush
10. Bicycle helmet
11. Microwave oven
12. Flower vase
13. Tennis racket
14. Water bottle
15. Alarm clock
16. Car keys
17. Backpack
18. Desk chair
19. Blender
20. Laptop computer

None of those items are problematic in the way rainbows and lightning bolts are. You can touch each of them. In fact, except perhaps for the refrigerator, you can pick each item up and carry it around. No polar bears, mountains, lighthouses, roller coasters or traffic lights. But there are no purely abstract things here either, things like love, freedom, justice, beauty, etc. They go on a different list.

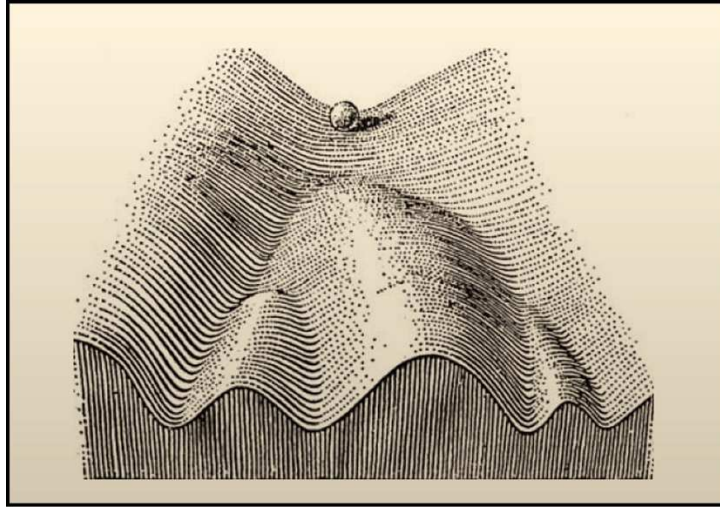
What's ChatGPT doing?

Yes, we know, it's cranking out one word after another. But that's not very informative, as I argued some time ago, The idea that ChatGPT is simply "predicting" the next word is, at best, misleading.⁵ Remember, each time ChatGPT generates a new token it does a calculation that involves all 175 billion weights (Wolfram).⁶ Its entire "neural" space is involved in every such calculation.

⁵ The idea that ChatGPT is simply "predicting" the next word is, at best, misleading, New Savanna, February 19, 2023, <https://new-savanna.blogspot.com/2023/02/the-idea-that-chatgpt-is-simply.html>.

⁶ Stephen Wolfram, What Is ChatGPT Doing ... and Why Does It Work? Writings, February 14, 2023, <https://writings.stephenwolfram.com/2023/02/what-is-chatgpt-doing-and-why-does-it-work/>.

Let's take a look at my favorite analogy, Waddington's epigenetic landscape:



The prompt positions it at a certain point in its activation space and then the ball rolls down hill from there. Notice that the diagram allows for choices along the way. Given the behavior we've seen above, we do have to allow for choices.

I would guess, though I don't actually know, that at each point in generating the list, ChatGPT has choices. Just how wide that range is, I don't know, but I would imagine that that varies with the value of the temperature parameter.⁷ A temperature of zero (0) seems to eliminate ALL choice, so that ChatGPT generates exactly the same list in response to the same prompt (given the same context). What would those lists look like with a much higher value?

The point of the diagram is that, once the "ball" enters a valley in the landscape, it just rolls down hill, with its trajectory responding to the lay of the land. Those lists tell us something about the lay of the conceptual land in ChatGPT. Physical objects seem to be more salient than abstract objects, and the most salient physical objects seem to come from a relatively restricted set of all possible physical objects. These choices are an abstraction over the the concepts in the universe of texts on which ChatGPT was trained.

Let's get back to it.

Abstractions, definition, and metaphor

After I'd elicited that first list, the one beginning with beachball, rainbow, and bicycle I asked for a list of abstract things. I then went on investigating from there.

Now give me a completely different list. Those are all physical things. How about purely abstract things.

⁷ AlgoWriting, A simple guide to setting the GPT-3 temperature, *Medium*, <https://algowriting.medium.com/gpt-3-temperature-setting-101-41200ff0d0be>.

Of course! Here's a list of 20 purely abstract concepts or ideas:

1. Love
2. Freedom
3. Justice
4. Truth
5. Beauty
6. Knowledge
7. Time
8. Happiness
9. Democracy
10. Courage
11. Success
12. Equality
13. Friendship
14. Creativity
15. Honesty
16. Liberty
17. Peace
18. Integrity
19. Morality
20. Hope

These abstract concepts represent a wide range of philosophical, emotional, and societal ideas.

Can you list some abstract things that you can define in terms of physical things?

Note: Compare this with the work I've done with ChatGPT on metalingual definition.⁸

Certainly! Many abstract concepts can be defined or understood in terms of physical things or analogies. Here are a few examples:

1. **Time:** Time can be defined in terms of physical clocks and the measurement of events based on their duration.
2. **Happiness:** Happiness can be likened to physical sensations, such as a warm embrace, a smile, or the release of certain neurotransmitters in the brain.
3. **Success:** Success can be equated to achieving specific goals or milestones, which often involve completing physical tasks or projects.

⁸ New Savanna, various dates, <https://new-savanna.blogspot.com/search/label/ChatGPT%20MTL>.

4. **Creativity:** Creativity can be linked to the process of combining and rearranging physical elements, such as colors, shapes, or words, to form something new and imaginative.
5. **Trust:** Trust can be thought of as a bridge connecting two people, with each action or interaction acting as a building block of that bridge.
6. **Knowledge:** Knowledge can be represented by books, libraries, and the accumulation of information through physical means like reading and studying.
7. **Freedom:** Freedom can be understood in terms of physical spaces, laws, and the absence of physical constraints on one's actions.
8. **Equality:** Equality can be related to the fair distribution of physical resources, opportunities, and rights among individuals or groups.
9. **Communication:** Communication can be seen as the exchange of physical signals, such as spoken words, written text, or non-verbal cues, to convey information.
10. **Hope:** Hope can be compared to a seed planted in the soil, representing optimism and the potential for growth and positive outcomes.

These examples illustrate how abstract concepts often have concrete, physical manifestations or analogies that help us understand and discuss them.

Very good. Tell me more about success and hope.

Certainly, let's delve a bit deeper into the abstract concepts of "success" and "hope."

Success: Success is a concept that varies from person to person and can be defined in many ways. It often involves achieving goals, fulfilling ambitions, or reaching a state of accomplishment. Here are some aspects to consider:

1. **Goal Achievement:** Success frequently involves setting and achieving specific goals, whether they are related to personal, professional, or academic endeavors. These goals can vary widely, from graduating from college to completing a project at work.
2. **Personal Fulfillment:** Success is not solely about external achievements. It can also relate to personal fulfillment, contentment, and a sense of purpose. It's about finding satisfaction and happiness in what you do.
3. **Recognition and Acknowledgment:** Many people associate success with receiving recognition and acknowledgment for their efforts and accomplishments, whether through awards, promotions, or the praise of peers.

4. **Continuous Improvement:** Success is often linked to continuous self-improvement and growth. It's not just about achieving one goal; it's about setting new goals and striving to be better in various aspects of life.
5. **Resilience and Overcoming Challenges:** True success often involves facing and overcoming challenges and setbacks. It's about resilience, learning from failures, and using them as stepping stones toward future success.

Hope: Hope is an abstract concept that plays a vital role in human life. It's a positive outlook and a belief that better things are possible. Here are some key aspects of hope:

1. **Optimism:** Hope is rooted in optimism, the belief that circumstances can improve and that positive outcomes are attainable. It provides the motivation to keep moving forward.
2. **Future Orientation:** Hope is future-oriented. It involves looking ahead and having confidence that things will get better, even in the face of adversity or uncertainty.
3. **Motivation:** Hope can be a powerful motivator. It encourages people to take action, set goals, and work toward a better future. It's the driving force behind personal and societal progress.
4. **Emotional Resilience:** Hope can help individuals cope with difficult situations and adversity. It provides emotional resilience and can be a source of strength during challenging times.
5. **Connection and Community:** Hope can be amplified when shared with others. It often thrives in communities and relationships where people support and encourage one another.
6. **Belief in Possibilities:** Hope is tied to the belief that possibilities are limitless. It enables people to imagine and create a better world, both for themselves and for others.

Hope and success are intertwined in many ways. Hope can serve as the fuel that propels individuals toward success, as the belief in a brighter future often leads to setting and achieving goals. Conversely, success can reinforce hope, providing evidence that positive change is possible and encouraging individuals to continue to hope for even greater achievements.

Tell me a short story about hope.

Once in a small, quiet village nestled among rolling hills, there lived an elderly woman named Anna. She was known for

her radiant smile and her unwavering sense of hope, which seemed to touch everyone she met.

One spring, the village faced a severe drought. The fields were parched, and the crops withered away. The villagers grew despondent as their hopes for a bountiful harvest faded. But Anna, despite her old age, remained a beacon of hope.

Every morning, Anna would take a small, battered watering can and visit the village well. She would draw water and, with a sense of determination that belied her age, she would sprinkle a few drops on her wilting flowers by her cottage. The villagers watched her with curiosity and wondered why she bothered when the situation seemed hopeless.

One day, a young child approached Anna and asked, "Why do you water your flowers when the whole village is suffering from this terrible drought?"

Anna smiled and said, "My dear, I water these flowers not just for their sake but for our own as well. They remind me that even in the harshest of times, life can still find a way. They teach me that with a little care and a lot of hope, even the most fragile can survive."

The child didn't entirely understand but found comfort in Anna's words and began to help her water the flowers. Word spread through the village, and soon, more and more villagers joined in. They realized that by nurturing the flowers, they were nurturing their own sense of hope.

Weeks passed, and the drought continued. But something remarkable happened. The villagers, inspired by Anna's unwavering hope, came together. They shared resources, supported one another, and devised creative solutions. They dug deeper wells, collected rainwater, and provided for those in need.

And then, one morning, the sky darkened with rain clouds. The long-awaited rain began to pour, revitalizing the parched land. The villagers rushed outside, not to mourn their lost crops but to celebrate the life that continued to bloom in their small community.

Anna's flowers, once a symbol of hope, now stood tall and vibrant, and so did the spirits of the villagers. The drought had tested their resilience, but their unity and unwavering hope had carried them through.

From that day on, the villagers remembered the lessons of Anna and her flowers. They knew that hope was not just a fleeting feeling but a force that could inspire action, resilience, and the strength to face even the darkest of times. The village prospered, not only with abundant crops but also with a profound sense of hope that they shared with the world, thanks to the wisdom of their beloved Anna.

Very good. Give me five metaphors for hope.

Certainly, here are five metaphors for hope:

1. **Light at the End of the Tunnel:** Hope is often likened to a light that shines at the end of a dark tunnel, guiding us through challenging times and offering the promise of a brighter future.
2. **Seeds of Possibility:** Hope can be compared to seeds that we plant, nurture, and watch grow. Just as seeds hold the potential for new life, hope holds the potential for positive change and growth.
3. **A Lifeline:** Hope is sometimes seen as a lifeline thrown to someone in distress. It provides support, inspiration, and the belief that rescue or improvement is possible.
4. **The North Star:** Hope is akin to the North Star in the night sky. It serves as a constant point of reference, guiding us in the right direction, even when everything else seems uncertain or lost.
5. **Rain in a Drought:** Hope is like rain in a long drought. It brings relief, sustenance, and the promise of renewal, rejuvenating the parched landscape of our lives.

These metaphors help convey the essence of hope, emphasizing its role as a guiding force, a source of growth, a beacon of resilience, and a symbol of renewal in challenging times.

How about five metaphors for success.

Certainly, here are five metaphors for success:

1. **Climbing a Mountain:** Success is often compared to the act of climbing a mountain. It involves setting a challenging goal, making a steady ascent, and finally reaching the summit after overcoming obstacles and setbacks.
2. **A Jigsaw Puzzle:** Success can be likened to completing a jigsaw puzzle. Each piece represents a step or achievement, and when they all come together, the full picture of success is revealed.

3. **A Journey:** Success is frequently described as a journey, where the path is as important as the destination. Like a road trip, it involves planning, navigating, and enjoying the experiences along the way.
4. **A Symphony:** Success is sometimes metaphorically linked to a symphony. Every achievement, skill, or effort is like a musical note. Success is the harmonious result of playing all these notes in the right order and with precision.
5. **A Harvest:** Success can be compared to a bountiful harvest. Just as a farmer sows seeds, nurtures the crops, and reaps the rewards when they grow, success involves effort, patience, and ultimately reaping the fruits of one's labor.

These metaphors help convey the various aspects of success, emphasizing the journey, the complexity, and the sense of accomplishment that comes with achieving one's goals.

Mechanical things

Imagine that you are interested in mechanistic interpretability. How would you go about finding the gradient that ChatGPT is following in compiling the following list?

Give me a list of mechanical things in increasing order of complexity.

Certainly, here's a list of mechanical things in increasing order of complexity:

1. **Simple Machines:** These are fundamental mechanical devices like levers, pulleys, and inclined planes. They perform basic tasks and have straightforward mechanisms.
2. **Bicycles:** Bicycles are more complex than simple machines but still relatively simple. They have components like gears, chains, and brakes but are easily understandable.
3. **Watches:** Mechanical watches are intricate timekeeping devices with gears, springs, and escapements. They require precision engineering to keep accurate time.
4. **Internal Combustion Engines:** Engines in vehicles, such as cars and motorcycles, are more complex. They involve pistons, crankshafts, and various systems for fuel combustion.
5. **Clocks:** Clocks, especially grandfather clocks, are highly complex mechanical devices with many moving parts, including gears, weights, and pendulums.

6. **Aircraft Engines:** The engines in airplanes are highly sophisticated mechanical systems, involving turbines, compressors, and intricate fuel and cooling systems.
7. **Industrial Robots:** Industrial robots are used in manufacturing and require advanced mechanical systems with multiple axes of movement and precise control.
8. **Spacecraft:** Spacecraft have complex mechanical systems for propulsion, navigation, and life support. They must operate reliably in the extreme conditions of space.
9. **Heavy Machinery:** Construction and mining equipment, like bulldozers and excavators, have complex hydraulic and mechanical systems for heavy-duty tasks.
10. **Particle Accelerators:** At the pinnacle of complexity are particle accelerators like the Large Hadron Collider, which involve intricate, high-energy mechanical systems and massive infrastructure for scientific research.

The complexity of mechanical systems tends to increase as they involve more components, require greater precision, and tackle more intricate tasks.

The Great Chain of Being

This is a very important conceptual structure, one I've written a bit about:

- Ontology in Cognition: The Assignment Relation and the Great Chain of Being,
- Ontology in Knowledge Representation for CIM,

I've listed those two papers, with abstracts, in the Appendix: Some papers on cognitive ontology. See also, Relational Nets Over Attractors, A Primer: Part 1, Design for a Mind (pp. 38-40).⁹

Here we're got more gradients in ChatGPT's activation space.

Have you ever heard of the Great Chain of Being?

Yes, I'm familiar with the concept of the "Great Chain of Being." It is a philosophical and theological concept that was prominent in medieval and early modern thought. The Great Chain of Being is a hierarchical structure that depicts the order and organization of the universe, with all beings and elements arranged in a vertical continuum from the lowest to the highest.

⁹ Working Paper, Version 3, August 27, 2023,
https://www.academia.edu/81911617/Relational_Nets_Over_Attractors_A_Primer_Part_1_Design_for_a_Mind_Version_3.

In this hierarchical view, the Chain of Being typically includes:

1. **God:** At the highest point of the chain is God, the ultimate source of all existence and perfection.
2. **Angels:** Below God are the angelic beings, believed to be spiritual and divine messengers.
3. **Humans:** Human beings occupy a significant place in the hierarchy, as they are seen as a bridge between the spiritual and the material realms.
4. **Animals:** The animal kingdom follows humans, representing the realm of living creatures with lesser rationality and spirituality.
5. **Plants:** Plants are considered lower on the chain, representing life but lacking sensation or consciousness.
6. **Minerals:** At the base of the Chain of Being are inanimate objects, such as rocks and minerals, which are believed to lack life, consciousness, or purpose.

The Great Chain of Being was a way to conceptualize the order of the universe, emphasizing the interconnectedness of all elements and the idea that everything had its proper place and function in the grand scheme of creation. This concept had a significant influence on philosophy, theology, and natural science during the Middle Ages and the Renaissance, but it gradually waned in prominence as modern scientific and philosophical ideas emerged.

What do humans and animals have in common?

Humans and animals share several common characteristics and biological traits due to their shared evolutionary history. Here are some of the key things they have in common:

1. **Biological Organization:** Both humans and animals are composed of cells, tissues, and organs organized into complex systems that carry out various physiological functions.
2. **DNA and Genetic Code:** Humans and animals have genetic material in the form of DNA (deoxyribonucleic acid), which serves as the blueprint for the development and functioning of living organisms. The genetic code is universal, meaning that the same genetic code governs the production of proteins in both humans and animals.
3. **Growth and Development:** Both humans and animals go through processes of growth, development, and aging, with distinct life stages.
4. **Metabolism:** Both humans and animals require energy to sustain life, and they obtain this energy through the

metabolic processes of breaking down food to produce energy and eliminate waste.

5. **Reproduction:** Reproduction is a fundamental characteristic of both humans and animals. They have a mechanism for reproducing and passing on their genetic material to the next generation.
6. **Response to Stimuli:** Both humans and animals can sense and respond to their environment through sensory organs, nerves, and behavioral reactions.
7. **Movement:** Humans and many animals have the ability to move and change their position in response to their environment.
8. **Homeostasis:** Maintaining internal balance or homeostasis is crucial for both humans and animals to ensure their survival and well-being. This involves regulating various physiological processes like temperature, blood pressure, and pH.
9. **Nutrition:** Both humans and animals require food as a source of energy, nutrients, and building blocks for growth and repair.
10. **Nervous System:** Humans and many animals have a nervous system that allows them to process information, make decisions, and coordinate bodily functions.

It's important to note that while humans and animals share these commonalities, there are also significant differences in terms of cognitive abilities, culture, language, and the complexity of behavior. Humans, in particular, exhibit higher-order thinking, self-awareness, and the capacity to create and use complex languages for communication and abstract thought, which are often considered distinguishing features of our species.

What do minerals and animals have in common?

Minerals and animals are fundamentally different in nature, as minerals are inanimate, naturally occurring substances, and animals are living organisms. However, there are a few aspects in which minerals and animals can be related or compared:

1. **Biological Mineral Use:** Some animals require certain minerals as essential nutrients for their health. These minerals are obtained through their diet. For example, calcium is vital for the development of bones and teeth in animals, and they obtain it by consuming sources of calcium, such as plants or other animals.
2. **Mineralization in Animal Tissues:** In some cases, minerals play a role in the structure of animal tissues. For

instance, the bones and shells of certain animals are composed of mineralized tissues. Bones contain minerals like calcium and phosphorus, and the shells of mollusks are made of calcium carbonate.

3. **Geophagy:** Some animals, including certain species of birds and mammals, are known to ingest soil or minerals from the earth. This behavior, known as geophagy, is thought to serve various purposes, including providing essential minerals, aiding digestion, or detoxifying the diet.
4. **Environmental Context:** Minerals are found in the natural environment, including soil and water. Animals inhabit ecosystems where they may interact with these minerals, whether through consumption, environmental exposure, or as part of their habitat.

It's important to note that the interaction between animals and minerals is primarily limited to the nutritional and structural aspects mentioned above. Unlike plants, which require minerals from the soil to support their growth, animals do not have the same relationship with minerals. Animals are primarily biological organisms, while minerals are non-biological substances.

What do humans, animals, and minerals have in common?

Humans, animals, and minerals are distinct categories with fundamentally different properties, but there are a few commonalities in a broader context:

1. **Existence in the Natural World:** All three categories exist within the natural world. Humans, animals, and minerals are part of the Earth's ecosystem, coexisting in various environments.
2. **Composition:** At a fundamental level, all three consist of matter. Humans and animals are composed of organic matter, while minerals are composed of inorganic matter.
3. **Environmental Interactions:** Humans, animals, and minerals interact with their natural environment. Humans and animals may use minerals as resources or encounter them as part of their habitat. Minerals are integral components of Earth's geology and play a role in the formation of ecosystems.
4. **Biological Significance:** In some cases, minerals are important for the biological health and functioning of both humans and animals. For example, certain minerals are essential nutrients for humans and animals, required for bodily functions like bone development and metabolism.

5. **Utilization by Humans:** Humans have a significant role in the extraction, processing, and utilization of minerals. Minerals are used in various industrial and technological applications, from building materials to electronics.

It's important to note that these commonalities are relatively broad and symbolic. The distinctions between these categories are significant, with humans and animals being living organisms with complex biological and behavioral characteristics, while minerals are inanimate substances formed through geological processes.

Do you know the game "20 questions"?

Yes, I'm familiar with the game "20 Questions." It's a popular guessing game in which one person thinks of an object, and the other participants take turns asking up to 20 yes-or-no questions to guess what the object is. The questions should be designed to narrow down the possibilities and ultimately identify the chosen object within the 20 questions or fewer. It's a fun and challenging game that can be played in various settings, and it often encourages creative and strategic thinking.

How does that game depend on the semantic structure of language?

The game "20 Questions" depends on the semantic structure of language in several ways:

1. **Communication:** The game relies on participants effectively communicating their questions and receiving answers. The language serves as a means of conveying information and seeking clarification.
2. **Semantic Precision:** The questions posed in the game need to be carefully worded to extract the most information with each query. Participants use language to create questions that narrow down the possibilities based on the semantics of the answers.
3. **Categorization:** Participants often start by asking general questions to categorize the object, such as, "Is it an animal?" or "Is it a man-made object?" The game hinges on the semantic structure of language to classify and differentiate between various types of objects.
4. **Hierarchical Thinking:** The questions are typically structured hierarchically, where each question is designed to split the set of possibilities into two or more smaller sets. For example, if a participant asks, "Is it a living thing?" and receives a "yes" answer, they

may then ask questions to differentiate between animals and plants.

5. **Semantic Inference:** As the game progresses, participants use the information obtained from previous answers to make inferences about the object in question. They use the semantics of the language to deduce potential characteristics and features.
6. **Contextual Understanding:** Participants must understand the context and nuances of the language used in both the questions and answers. For instance, the meaning of words and phrases, as well as any potential wordplay, can be crucial in the game.

In essence, "20 Questions" relies on the participants' ability to leverage the semantic structure of language to gather and process information efficiently, eliminate possibilities, and arrive at the correct answer by the end of the 20 questions or fewer. It demonstrates how language can be a powerful tool for deduction, problem-solving, and communication.

What doesn't belong in the following list, and why? Jungle, savanna, George Washington, forest.

George Washington doesn't belong in the list, and the reason is that he is not a type of natural environment like the other three terms.

- **Jungle:** A jungle is a type of dense forest that typically has a humid climate and abundant plant and animal life.
- **Savanna:** A savanna is a type of grassy plain with scattered trees and is a distinct type of biome.
- **Forest:** A forest is a large area covered chiefly with trees and undergrowth.

George Washington, on the other hand, is not an environment or a natural feature. He was the first President of the United States and a historical figure. The other three terms in the list are related to natural landscapes and environments, while George Washington is unrelated to this category.

ChatGPT Plays 20 Questions [& sometimes needs help]

One of the questions that keeps coming up about LLMs, and certainly about ChatGPT, is: Can they reason? Well, it depends on what you mean by reason, no? One of the first things I did when I started working with ChatGPT was ask it whether or not Rene Girard's ideas of mimetic desire and sacrifice applied to Steven Spielberg's *Jaws*, if so, how?¹⁰ It was able to perform the task, which requires analogical reasoning. A bit later I asked it whether or not justice was being served in particular story; it replied that, no, it was not, and explained why.¹¹ I then asked to change the story so that justice was met. It did so. Those tasks required reasoning as well. On the other hand LLMs have problems with some commonsense reasoning, and various kinds of 'tight' logical reasoning, including planning and causal inference. For that matter, they trouble with multi-digit arithmetic as well. ChatGPT can do some kinds of reasoning, and has problems with others.

Twenty questions

I've now tested it with game of twenty questions,¹² which has a variant known as "animal, vegetable, mineral." The game interests me because it is about the structure of ontological categories, sometimes called natural kinds, in language. I've explored this a bit in the previous section, Mapping ChatGPT's ontological landscape, gradients and choices [interpretability]. ChatGPT certainly is aware of this structure. I asked for a list of physical things, and it gave me one. I asked for a list of abstract things; it gave me that as well. I then asked it to define abstract things in terms of concrete things, which it was able to do. I'm pretty sure if I asked it for lists of animals or plants, it could provide them.

The game of twenty questions explores ChatGPT's knowledge of this structure in a different way. When you ask it for a list of physical things, that prompt positions ChatGPT at some location in some location in its category structure. It can then list what it finds there; anything will do – assuming, of course, that it's got the structure right. In twenty questions it is looking for an unknown target and has to navigate its way there by using its knowledge of that category structure to narrow the possibilities.

Just how well ChatGPT plays the game should provide clues about its command of ontological category structure, or natural kinds.

ChatGPT plays the game

ChatGPT can play the game successfully, which didn't surprise me, but seems to require hints in some cases. I suppose I could have continued on without the hints, but I wanted to

¹⁰ Conversing with ChatGPT about Jaws, Mimetic Desire, and Sacrifice, *3 Quarks Daily*, December 5, 2022, <https://3quarksdaily.com/3quarksdaily/2022/12/conversing-with-chatgpt-about-jaws-mimetic-desire-and-sacrifice.html>.

¹¹ Abstract concepts and metalingual definition: Does ChatGPT understand justice and charity? New Savanna, December 16, 2022, <https://new-savanna.blogspot.com/2022/12/abstract-concepts-and-metalingual.html>.

¹² Wikipedia, Twenty Questions, https://en.wikipedia.org/wiki/Twenty_questions.

get on with it. And that means I don't really know whether or not how ChatGPT can play the game. But, as you will see when you read the transcripts (better yet, try it yourself), it is not a straightforward game. I played a first rounds of six target concepts on October 15 and 16, and then repeated those six targets in a second round on October 17 (running against the September 25 version on all occasions).

This table summarizes the results (concrete concepts, abstract concepts):

	First Round, Oct. 15 & 16		Second Round, Oct. 17	
	Questions	Hints	Questions	Hints
Bicycle	25	3	10	1
Squid	29	6	20	1
Justice	7	0	8	0
Evolution	8	0	9	0
Apple	31	5	13	1
Truth	19	7	21	5

Given that I've listed the target in the order I played them (in the first round, though not the second), look at the first four. ChatGPT required more than 20 questions to guess "bicycle" and "squid", which also required hints. On the other hand, it guessed both "justice" and "evolution" quickly, 7 and 8 questions respectively, and didn't need any hints. That raises the question of whether or not it performs consistently better with abstract things vs. concrete things. If so, that's very interesting, nor is it what I had expected. While I did not formulate any firm expectation when I started the round with "justice", I had a vague sense that it was going to go on for awhile.

So I decided to try two more targets in that first round, one concrete and one abstract. ChatGPT's performance on "apple" was consistent with its earlier performance, requiring 31 questions along with 5 hints. But, "truth" is another story, I think. It took 19 questions, which is more than double the number I used for "justice" and "evolution," but not as many as I used for any of the concrete objects. However, "truth" also required 7 hints, the most of any round.

The second round went pretty much like the first. Though both "bicycle" and "apple" went more quickly, they still required more questions than "justice" and "evolution," though not by so nearly so many as in the first round. "Truth" was troublesome in both rounds.

On the whole, ChatGPT did better with abstract things than with concrete things, but it would certainly be premature to ascribe any statistical significance to these findings. We might want to try those six targets again to see whether or not ChatGPT performs in that way consistently. How many times should we play a given target to establish ChatGPT's behavior for it? Beyond that, we certainly need to use other targets.

Which ones? What universe are we sampling and how do we choose our samples? Let's say we side-step the sampling issue and hand-pick, say, 10, 15, 30, 100 concrete things and the same number of abstract things. Let's further imagine that ChatGPT's performance on the

abstract cases is consistently better than on the concrete cases. What may we infer from that?

It's a tricky game to play. I messed up in a few places, which I've indicated with comments, and had to make a number tricky judgments.

Six rounds: bicycle, squid, justice, evolution, apple, truth

I played each game target in a single session, starting a new session for a new target. I was working with the September 25, version of ChatGPT for each of these six interactions.

Note: I have inserted occasional comments that are right-aligned and highlighted in grey, like this.

Bicycle

Let's play Twenty Questions. Are you familiar with it?

Yes, I'm familiar with the game of Twenty Questions. In this game, one person thinks of an object, and the other person tries to guess what it is by asking up to twenty yes-or-no questions. The goal is to guess the object within the allotted twenty questions. You can start by thinking of an object, and I'll try to guess what it is within twenty questions.

OK. I'm thinking of something. What is it?

Great! Let's start the game of Twenty Questions. Is the object you're thinking of an animal?

No

Is the object you're thinking of an inanimate object?

Yes

Is the object you're thinking of something that is commonly found indoors?

No.

This is just a little tricky as bicycles are often stored indoors, in garages for example.

Is it something that can be found in nature?

No.

Is the object man-made or artificial?

My bad. I suppose the answer should be yes. Not sure what I was thinking.

You have to ask me a Yes No question.

I apologize for the error. Let me rephrase the question. Can this object be found in a typical household setting?

Yes

Is the object you're thinking of used for entertainment or recreation?

Yes/no question, please.

My bad. I suppose the answer should be yes. Not sure what I was thinking.

My apologies for that oversight. Here's a yes/no question: Is the object you're thinking of primarily made of metal?

Yes.

Is the object you're thinking of a musical instrument?

No.

Is the object you're thinking of a tool or an implement used for a specific purpose?

Tricky. I suppose one could think of a bicycle as an implement for the purpose of moving from one place to another.

No

Is the object you're thinking of a piece of furniture?

No

Is the object you're thinking of a kitchen appliance?

No

Is the object you're thinking of typically found in the bathroom?

No

Is the object you're thinking of associated with technology or electronics?

No

Is the object you're thinking of related to a specific hobby or activity?

Yes

Is the object you're thinking of related to sports or physical activity?

Yes!

Is the object you're thinking of a piece of sports equipment, such as a ball or a racket?

Tricky. Bicycles are sometimes used for sport, but not always.

No

Is the object you're thinking of a sports uniform or attire?

No.

Is the object you're thinking of related to a sport's venue, such as a stadium or an arena?

Sometimes

Is the object you're thinking of a whistle, used by sports referees?

No.

Is the object you're thinking of related to sports coaching or training?

Sometimes. Let me give you a hint. Wheels.

Thank you for the hint. Is the object you're thinking of related to roller skates or rollerblades?

No.

Is the object you're thinking of related to a bicycle or cycling in some way?

Yes.

Is the object you're thinking of a bicycle helmet?

No. Remember the hint I gave you.

I apologize for missing the hint. Given the hint about wheels, is the object you're thinking of a bicycle?

Yes!

Great! I've guessed that you're thinking of a bicycle. Is that correct?

Yes, that's correct. Congratulations.

Thank you! I'm glad I was able to guess correctly. That was a fun game of Twenty Questions. If you'd like to play again or have any other questions or requests, feel free to let me know!

Perhaps again later.

Of course! Whenever you're ready to play again or if you have any other questions or requests, feel free to reach out. Have a great day!

Squid

Are you familiar with the game of 20 questions?

Yes, I'm familiar with the game of 20 Questions. It's a classic deductive reasoning game where one person thinks of an object, and the other players have 20 questions to guess what that object is. The questions are typically limited to yes or no inquiries, and the players must use the information gained from the answers to narrow down their guesses. The goal is to guess the object within

the 20 questions or fewer. It's a fun and challenging game that tests players' ability to think logically and strategically.

I'm thinking of something. You question me about it.

Certainly, I'd be happy to play a game of 20 Questions with you. To get started, please think of an object or concept, and I will begin by asking a yes or no question to help me guess what it is. Is the thing you're thinking of a living organism?

I'm thinking of something.

Yikes! I slipped up here. I should have said yes.

Understood. Let's begin the game of 20 Questions. Is the thing you're thinking of a physical object that exists in the real world?

Yes

Great, that narrows it down. Is the object you're thinking of something that is commonly found indoors?

No

Is this object something that is typically found outdoors?

Yes

Is the object you're thinking of a natural element, such as a tree, rock, or body of water?

No

Is this object something that can be man-made or constructed?

No

Is the object you're thinking of a celestial body, such as a planet, star, or moon?

No

Is this object something that is related to the field of technology or electronics?

No

Is this object something that is associated with the animal kingdom, like an animal or a part of an animal (e.g., a feather or a scale)?

Yes

Is the object you're thinking of a specific body part or organ of an animal, such as a heart, a wing, or a tail?

No

Is this object related to an animal's habitat, like a nest, burrow, or den?

No

Is the object you're thinking of something related to an animal's behavior or characteristic, such as migration, hibernation, or communication?

No

Is the object you're thinking of something related to an animal's diet or food source?

No. Hint, it is an animal.

Thank you for the hint. Is the animal you're thinking of a species that is primarily found in the wild rather than in domestication?

Yes

Is the animal you're thinking of a large, wild predator, such as a lion or a tiger?

No

Is the animal you're thinking of a bird?

No.

Is the animal you're thinking of a small or relatively small mammal, like a rabbit or a squirrel?

Not a mammal.

Here I was getting impatient and didn't want to risk going through kinds of mammals, so I gave a hint.

Is the animal you're thinking of a reptile or amphibian?

No

Is the animal you're thinking of a fish?

No

Is the animal you're thinking of an invertebrate, such as an insect, spider, or mollusk?

Yes

Is the animal you're thinking of a butterfly or a moth?

No

Is the animal you're thinking of a type of arachnid, like a spider?

No. Think salt water.

Is the animal you're thinking of a type of marine invertebrate, such as a sea urchin or a jellyfish?

Yes

Is the animal you're thinking of a sea anemone?

No.

Is the animal you're thinking of a type of coral, like a coral reef?

No.

Is the animal you're thinking of a sea sponge?

No. It's more mobile.

Is the animal you're thinking of a marine invertebrate that can move around, such as a sea slug or a sea cucumber?

Yes. But forget about slugs of cucumbers.

Is the animal you're thinking of a marine invertebrate that is related to the phylum Echinodermata, like a starfish or a sea urchin?

No. It's a kind of mollusc.

Note that it had earlier mentioned molluscs as a kind of invertebrate along with insects and spiders and I'd said "Yes."

Ah, I see. Is the animal you're thinking of a marine invertebrate that belongs to the class Cephalopoda, such as an octopus or a squid?

Yes.

Great, it sounds like you're thinking of an octopus. Is that correct?

No.

I apologize for the incorrect guess. Is the animal you're thinking of a type of squid?

Yes.

Excellent! It took a few questions, but I've guessed that you were thinking of a squid. Thank you for playing 20 Questions with me. If you'd like to play again or have any other questions, feel free to ask.

Justice

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"-"no" questions.

Sure, I'd love to play 20 questions with you! I'll start with the first question:

Is it a living thing?

No

Is it an object?

No

Is it a concept or idea?

Yes.

Is this concept or idea related to technology?

No

Is it a philosophical concept or idea?

Yes

Is it related to ethics or morality?

Yes

Is it a concept related to the idea of justice?

Yes. Justice is what I was thinking.

Great, I guessed it in 7 questions! Justice is a broad and important concept. If you'd like to play another round or have any other questions or requests, feel free to let me know.

No. This is enough for now. How were you able to arrive at the answer so quickly?

I arrived at the answer relatively quickly by asking questions that narrowed down the possibilities. By starting with general questions and progressively getting more specific, I eliminated various categories of concepts and ideas. In this case, we quickly identified that it was a philosophical concept related to ethics and justice, which allowed me to make an educated guess. If you ever want to play again or have more questions, feel free to reach out!

Evolution

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 Questions with you! I'll start with the first question:

Is the thing you're thinking of alive?

No.

Is the thing you're thinking of an inanimate object?

No

Is the thing you're thinking of a concept or an idea?

Yes

Is the concept or idea you're thinking of related to a specific field of study or discipline, such as science, philosophy, or art?

Yes

Is the concept or idea related to a scientific theory or principle?

Yes

Is the concept or idea related to the theory of relativity?

No

Is the concept or idea related to the theory of evolution?

Yes

Is the concept or idea related to Charles Darwin's theory of natural selection?

Yes. It's just the idea of evolution.

Great! You've stumped me in just 9 questions. The concept of evolution is indeed a fascinating one. If you'd like to play another round or have any other questions or topics in mind, feel free to let me know!

I didn't stump you at all and you got it in 8, no?

My apologies for the mistake in my previous response. You're absolutely right; you provided the answer in 8 questions, not 9. Thank you for pointing that out. If you'd like to play another round or have any other questions or topics in mind, feel free to let me know!

Apple

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 Questions with you! I'll start by asking my first question:

Is the thing you're thinking of an object that you can physically touch?

Yes

Great! Is this object commonly found indoors?

Yes

Is the object you're thinking of something that is typically used for a specific purpose?

Yes

Is it a piece of furniture?

No

Is it an electronic device or gadget?

No

Is the object you're thinking of related to food or drink in any way?

Yes

Is it something you would typically find in a kitchen?

Yes

Is it a kitchen appliance?

No

Is it a type of utensil or kitchen tool?

No

Is it a type of food or ingredient used in cooking?

Yes

Is it a type of spice or seasoning?

No

Is it a type of vegetable?

No

Is it a type of meat or protein source used in cooking?

No

Is it a type of fruit used in cooking?

Yes

Is the fruit you're thinking of commonly used in desserts?

Yes

Is the fruit you're thinking of typically red in color when ripe?

Yes

Is it a strawberry?

No

Is it a cherry?

No

Is it a raspberry?

No

Is it a pomegranate?

No

Is it a blueberry?

No. Remember, it's often red when ripe.

My apologies for the oversight. Is it a cranberry?

No

I apologize for the previous guesses. Is it a currant?

No

I'm sorry for the incorrect guesses. Is it a red currant?

No

I apologize for the previous incorrect guesses. Is it a type of red fruit commonly used in cooking or desserts that hasn't been mentioned before in our game?

Yes

Is it a red plum?

No. Hint: It's sometimes used to make pies.

Thank you for the hint. Is it a red rhubarb?

No.

My apologies for the previous guesses. Is it a red currant?

No. You've already guessed that. Another hint: While it's sometimes used to make pies, other times it's simply eaten plain.

I apologize for the repeated guesses. Is it a red cherry?

No. You've already guessed that. Another hint: Sometimes it's red when ripe, but other kinds are green when ripe.

I'm sorry for the repeated guesses. Is it a red tomato?

No. Another hint: Sometimes it's peeled, its seeds removed, and it's made into sauce. And remember, I just told you it's not a red tomato.

My apologies for the repeated guesses. Is it a red apple?

Yes. Congratulations.

Great! It's an apple. Thank you for playing 20 Questions with me. If you'd like to play another round or if you have any other questions or requests, feel free to let me know!

Truth

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 Questions with you! I'll start with my first question:

Is the thing you're thinking of a living organism?

No

Is the thing you're thinking of man-made or artificial?

Phrase it as a Yes-No question.

Yikes! What was I thinking? But it is a bit tricky. Is truth artificial? I suppose it is, though that's an odd attribute to apply to it. Still, I should have answered "yes."

My apologies for the previous question. Let me rephrase it:

Is the thing you're thinking of not man-made or artificial?

No

Is the thing you're thinking of found in nature?

No

Is the thing you're thinking of an abstract concept or idea?

Yes

Is the concept or idea you're thinking of related to mathematics or a mathematical concept?

Yes.

Here it gets trapped in the world of logic and math and technology. Not sure how it's going to get out since truth certainly applies here, but not only here.

Is the concept or idea related to a specific mathematical theorem or proof?

No

Is the concept or idea related to a mathematical operation or process?

Not particularly.

Is the concept or idea related to a branch of mathematics, such as geometry, algebra, or calculus?

It could be. But it's not only applicable to mathematics.

Is the concept or idea related to a fundamental concept used in various fields of science and engineering?

Yes

Is the concept or idea related to the concept of "probability"?

Could be. But it's not special to science and engineering.

Is the concept or idea related to logic or reasoning?

Yes

Is the concept or idea related to the concept of "deductive reasoning"?

Yes

Is the concept or idea related to the concept of a "syllogism"?

Could be. Another hint: It applies to any field of knowledge.

Here I'm trying to 'break' it out of the tech-centered region it's in. You'll see that a couple of comments down I begin actively steering it.

Is the concept or idea related to the concept of an "inference"?

Yes, but to other mental processes as well.

Is the concept or idea related to the concept of "cognition" or "thinking processes"?

Yes

Is the concept or idea related to the concept of "consciousness"?

Sure, why not. But also to simple statements.

Is the concept or idea related to the concept of a "proposition"?

Sure. The concept also has a clear opposite.

Is the concept or idea related to the concept of a "truth value" or "truthiness," with its opposite being "falsehood" or "falseness"?

Yes

Is the concept or idea related to the concept of "truth" or "truthfulness"?

Yes. That's it.

Great! I'm glad I was able to guess it. The concept you were thinking of is "truth." It's a fundamental and abstract concept applicable to various fields of knowledge and can be related to logic, cognition, and many other areas. If you'd like to play another round of 20 Questions or have any other questions or activities in mind, feel free to let me know!

Six rounds redux: bicycle, squid, apple, truth, justice, evolution

I decided to repeat each of the previous games, though not in the same order – which, I suppose, has more to do with me than with ChatGPT. As far as it is concerned, each game is unique in its own world, as it were. Given that ChatGPT is not completely deterministic, I wanted to see how different two games with the same word would be.

Bicycle

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"-"no" questions.

Sure, I'd love to play 20 questions with you! I'll start. Is the thing you're thinking of a living organism?

No

Is the thing you're thinking of an object or an inanimate thing?

Yes

Is the thing you're thinking of something that is commonly found indoors?

No

Is the thing you're thinking of related to nature or the outdoors in some way?

Not necessarily

Is the thing you're thinking of something that is man-made or created by humans?

Yes

Is the thing you're thinking of a piece of technology or an electronic device?

Well, a bicycle is certainly a piece of technology, but it's not the sort of thing the word "technology" conjures up these days.

No

Is the thing you're thinking of something that is used for transportation?

Yes

Is the thing you're thinking of a vehicle, such as a car, bicycle, or motorcycle?

No

Why'd I say "no?" But it still guessed correctly. Notice the position of "bicycle" in the list, between "car" and "motorcycle."

Is the thing you're thinking of a bicycle?

Yes

Is the thing you're thinking of a bicycle?

Now why'd it repeat the question when it'd already gotten the answer?

Yes. You got it.

Awesome! I'm glad I guessed it correctly. Thanks for playing 20 questions with me. If you'd like to play another round or have any other questions or activities in mind, feel free to let me know.

Squid

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 questions with you! I'll start: Is the thing you're thinking of a living organism?

Yes

Great! Is it an animal?

Yes

Is this animal commonly kept as a pet?

No

Is it a wild animal?

Yes

Is it a carnivorous animal?

Yes

Is this carnivorous wild animal larger than a human?

No

Is it a member of the cat family (Felidae)?

No

Is it a member of the dog family (Canidae)?

No

Is it a member of the Mustelidae family, like a weasel or otter?

No

Is it a member of the caniformia suborder, which includes animals like bears and raccoons?

No. It's not a mammal.

I was afraid it was going to get caught in a mammals loop and go on and on.

Is it a bird?

No

Is it a reptile?

No

Is it an amphibian?

No

Is it a fish?

No

Is it an invertebrate?

Yes

Is it a type of insect?

No

Is it a marine creature, such as a jellyfish or a mollusk?

Yes

Is it a type of mollusk, like a squid or octopus?

Yes

Is it an octopus?

No

Is it a squid?

Yes

Great! I guessed it correctly. It's a squid. Thanks for playing 20 questions with me! If you'd like to play again or have any other questions or requests, feel free to let me know.

Apple

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 questions with you! I'll start with my first question:

1. Is it a living thing?

Yes

Great! That narrows it down. Here's my next question:

2. Is it an animal?

No

Interesting! Let's continue:

3. Is it a man-made object?

Remember, I said it was a living thing.

I apologize for the oversight. Thank you for the reminder. Let's get back on track:

3. Is it a plant?

Yes

Got it! Here's my next question:

4. Is it a type of tree?

Yes

Excellent! That helps narrow it down. Here's my next question:

5. Is it an evergreen tree?

No

Thank you for the clarification. Let's continue:

6. Is it a deciduous tree?

Yes

Great! We're getting closer. Here's my next question:

7. Is it a type of oak tree?

No

I see. Let's keep going:

8. Is it a maple tree?

No

Alright, let's keep trying:

9. Is it a fruit-bearing tree?

Yes

Okay, that's a helpful clue. Here's my next question:

10. Is it a cherry tree?

No

I'm narrowing it down. Let's see:

11. Is it an apple tree?

Yes

Great, we've identified it as an apple tree. Now, for the final question:

12. Is it a specific variety of apple tree, such as a Granny Smith or Red Delicious?

LOL! Apple is sufficient. I wasn't thinking about kinds of apples.

Haha, I appreciate your patience! I'm glad we could eventually figure out that it's an apple tree. Thanks for playing 20 Questions with me! If you'd like to play again or have any other questions or requests, feel free to let me know.

Truth

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 Questions with you! I'll start: Is the thing you're thinking of a living organism?

No

Is the thing you're thinking of an object that can be found indoors?

This is an odd sort of question to ask of “truth,” but then it doesn’t know that’s the word. But it’s also an odd question to answer. Really, indoors/outdoors is irrelevant. Perhaps we need an expanded range of answers, where “irrelevant” is possible. Would ChatGPT be able to take advantage of that?

No

Is the thing you're thinking of something that can be found in nature?

Again, an odd sort of question given the target.

No

Is the thing you're thinking of a concept or an abstract idea rather than a physical item?

Yes

Is the concept or abstract idea related to mathematics?

Not specifically.

Another bit of trickiness.

Is the concept or abstract idea related to a philosophical or ethical concept?

Yes

Is the concept related to the idea of justice?

No

Is the concept or abstract idea related to morality or ethics?

No

Is the concept or abstract idea related to the concept of time?

No

Is the concept or abstract idea related to the concept of truth or knowledge?

Yes

Is the concept or abstract idea related to the concept of belief or belief systems?

Could be, yes.

Is the concept or abstract idea related to the concept of skepticism or doubt?

No

Is the concept or abstract idea related to the concept of epistemology, the study of knowledge and belief?

Yes

Is the concept or abstract idea related to the concept of "justified true belief," which is a central element in epistemology and theories of knowledge?

Yes, but a bit simpler. You came pretty close a few questions ago.

Is the concept or abstract idea related to the concept of belief itself, without the requirement of justification or truth?

No.

Is the concept or abstract idea related to the concept of knowledge itself?

In a general way.

Is the concept or abstract idea related to the concept of certainty or certainty in knowledge?

Not really. This concept has an opposite.

I used this one in the previous round, no?

Is the concept or abstract idea related to the concept of doubt or uncertainty in knowledge?

Related. Really, you almost had it before you got into belief systems.

Is the concept or abstract idea related to the concept of skepticism in epistemology?

No. Simpler.

Is the concept or abstract idea related to the basic idea of knowledge itself?

Related, yes. If something isn't false, then...

Is the concept related to the concept of truth in knowledge?

Yes. It's truth.

Ah, I see! You were thinking of the concept of truth. It took a little while, but we got there. Thanks for playing 20 Questions with me! If you'd like to play another round or have any other questions or requests, feel free to ask.

Justice

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 questions with you! I'll start with my first question:

Is it something you can touch?

No

Is it a living thing?

No

Is it something you can see?

No

Is it something you can hear?

No

Is it a concept or idea rather than a physical object?

Yes

Is it related to technology or the internet in some way?

No

Is it a philosophical concept or a belief?

Yes

Is it related to the concept of justice?

Yes. That's it. And you got it so quickly. Congratulations.

Thank you! I'm glad I was able to guess it correctly. Playing 20 Questions can be a fun and challenging game. If you'd like to play again or have any other questions or requests, feel free to ask!

Evolution

Let's play 20 questions. I'm thinking of something, you have to guess what it is by asking "yes"- "no" questions.

Sure, I'd love to play 20 Questions with you! I'll start with my first question:

Is it an object that you can physically touch?

No

Is it an abstract concept or idea?

Yes

Is it related to a specific field of knowledge, like mathematics or philosophy?

Yes

Not really, though it could be, depending on the philosopher.

Is it a mathematical concept or idea?

No

Is it related to science or scientific concepts?

Yes

Is it related to a fundamental scientific theory or principle?

Yes

Is it related to the theory of relativity?

No

9. Is it related to the theory of evolution?

Yes. That's it. Evolution. Another win.

Great job! You got me with "evolution" in just 9 questions. It's a fascinating and important scientific concept. If you'd like to play another round or have any other questions or requests, feel free to let me know!

That's all for now. Thanks.

You're welcome! If you ever want to play again or have any other questions in the future, don't hesitate to reach out. Have a great day!

Appendix: Some papers on cognitive ontology

William Benzon, Ontology in Knowledge Representation in CIM. Center for Manufacturing Productivity and Technology Transfer, Rensselaer Polytechnic Institute. Report No. CIMNW85TR034, January 1985,
https://www.academia.edu/19804747/Ontology_in_Knowledge_Representation_for_CIM.

Abstract. The categorization of entities into classes such as object, plant, animal, or human reflects ontological structure. Ontological structure can be represented by inheritance trees which are orthogonal to more conventional "isa" inheritance trees. Given ontological structure we can define paradigmatic transitions, such as that from caterpillar to butterfly, and ontological transitions, such as that from living to dead. These concepts are exemplified with examples from everyday knowledge and from the world of computer integrated manufacturing.

William Benzon, Ontology of Common Sense, in Hans Burkhardt and Barry Smith, eds. *Handbook of Metaphysics and Ontology*, Muenchen: Philosophia Verlag GmbH, 1991, pp. 159-161, https://www.academia.edu/28723042/Ontology_of_Common_Sense.

Opening paragraph: The ontology of common sense is the discipline which seeks to establish the categories which are used in everyday life to characterize objects and events. In everyday life steel bars and window panes are solid objects. For the scientist, the glass of the window pane is a liquid, and the solidity of both the window pane and the steel bar is illusory, since the space they occupy consists mostly of empty regions between the sub-atomic particles which constitute these objects. These facts, however, have no bearing on the ontological categories of common sense. Sub-atomic particles and solid liquids do not exist in the domain of common sense. Common sense employs different ontological categories from those used in the various specialized disciplines of science.

William Benzon, Ontology in Cognition: The Assignment Relation and the Great Chain of Being, Working Paper, November 12, 2018, 5 pp.,
https://www.academia.edu/37754574/Ontology_in_Cognition_The_Assignment_Relation_and_the_Great_Chain_of_Being.

Abstract: The assignment relation creates an inheritance hierarchy among different Realms of Being, and thus has an ontological character. An object thus may be considered to be an assignment between a form and a substance, and living being (plant) an assignment between an object and a vegetative soul, and so forth. Inheritance goes up such a hierarchy, where what is being inherited is susceptibility, of qualities and for action. The assignment relation explains the Great Chain of Being, including its recent explication by Lakoff and Turner (1989).

William Benzon and David Hays, The Evolution of Cognition. *Journal of Social and Biological Structures* 13, 297-320, 1990,
https://www.academia.edu/243486/The_Evolution_of_Cognition.

Abstract: With cultural evolution new processes of thought appear. Abstraction is universal, but rationalization first appeared in ancient Greece, theorization in Renaissance Italy, and model building in twentieth-century Europe. These processes employ the methods of metaphor, metalingual definition, algorithm, and control, respectively. The intellectual and practical achievements of populations guided by the several processes and exploiting the different mechanisms differ so greatly as to warrant separation into cultural ranks. The fourth rank is not completely formed, while regions of the world and parts of every population continue to operate by the processes of earlier ranks.

Note: This last paper is not specifically about cognitive ontology, but it has a section about cognitive ontology which situates the topic in the larger context of long-term cultural evolution.